DIPLOMA PROJECT — PRE-DEFENSE DOCUMENTATION

Development of an Air Quality Forecasting System Based on IoT and Machine Learning

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1. Project Proposal

1.1 Project Title and Team

Project title: Development of an Air Quality Forecasting System Based on IoT and Machine Learning.

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Supervisor: Marzhan Turarova (Turarova Marzhan Kabdykalievna).

1.2 Problem Statement

Air quality changes over time and may affect public health and everyday decisions. Monitoring systems describe current or past conditions, while users may also benefit from short-term forecasts. This project addresses the engineering problem of developing a system that receives air-quality-related data, processes it, applies a machine-learning model, and presents forecast results in an understandable form.

1.3 Relevance

Air pollution is a relevant environmental issue for urban areas, including Astana. IoT can support environmental data acquisition and transfer, while machine learning can identify patterns in historical observations and produce short-term forecasts. The project will investigate how these elements can be combined in a practical, testable prototype.

1.4 Target Users

- Residents interested in current and near-future air-quality conditions.
- Users interested in environmental monitoring and pollution trends.
- Researchers or students exploring IoT-based environmental data and forecasting.

1.5 Aim

To design and develop a prototype system for forecasting air quality using IoT-compatible data acquisition and machine-learning methods, and to evaluate its performance using defined metrics.

1.6 Objectives

- Review recent research on IoT-based air-quality monitoring and ML forecasting.
- Compare existing systems and alternative technologies relevant to the project.
- Define functional and non-functional requirements with acceptance criteria.
- Design the system architecture and required diagrams.
- Select and prepare a historical air-quality dataset and define an IoT-compatible ingestion interface.
- Establish a baseline and compare it with at least one ML model.
- Develop a prototype for data processing, forecasting, and presenting results.
- Test the prototype and evaluate forecast quality and relevant system performance.
- Document implementation decisions, limitations, and evaluation results.

1.7 Proposed Solution

The planned prototype consists of a data input layer, preprocessing, data storage, a forecasting component, and a results presentation layer. Initial model development and evaluation will use a suitable historical dataset. The input layer should allow later integration of physical IoT measurements if hardware is available; physical sensor deployment is not assumed as a completed capability.

1.8 Expected Outcomes

- Documented system design and prototype.
- Prepared dataset and reproducible preprocessing workflow.
- Baseline and ML forecasting model evaluated under comparable conditions.
- Results view showing forecast outputs and relevant air-quality information.
- Test/evaluation report stating metrics, limitations, and conditions.

1.9 Scope and Limitations

The scope includes requirements analysis, system design, data preparation, model comparison, prototype implementation, and evaluation. The exact forecast target (pollutant concentration or AQI), forecast horizon, dataset, model, and final stack must be fixed after checking data availability and requirements. Direct sensor connection depends on hardware availability. The prototype is not intended to replace certified measurement stations or provide medical advice.

Two candidate datasets are being considered: the AirData.kz Open Air Quality Dataset for Kazakhstan and the Beijing Multi-Site Air Quality Dataset from the UCI Machine Learning Repository. The Kazakhstan dataset is relevant to the project's intended geographical context, while the Beijing dataset provides a separate multi-station time-series dataset for methodological comparison. The final dataset selection will depend on data coverage, completeness, available variables, and suitability for the chosen forecasting target.

1.10 Project Management and Quality Assurance

Work should be tracked through a shared GitHub or GitLab repository and YouTrack. Tasks should have an owner, expected outcome, acceptance criteria, and deadline. Documentation, diagrams, and code should be committed progressively, with identifiable contributions from both students. Significant changes should be reviewed before merging.

2. Evidence of Relevance

2.1 Evidence Collection Method

The relevance of the proposed project was assessed through a review of publicly available reports and air-quality data sources related to Kazakhstan. This approach was selected instead of user interviews or client agreement. The review focused on the documented air-pollution problem, the availability of historical measurements, and the potential value of forecasting air quality.

2.2 Evidence Summary

Evidence source	Observation / finding	Implication for the project
World Bank, *Air Quality Management in Central Asia* (2025)	The report identifies air pollution as a significant environmental and public-health challenge in Central Asia. PM2.5 concentrations in major cities frequently exceed WHO guideline levels, particularly during winter.	Supports the relevance of developing tools that help users examine air-quality trends and forecast future pollution levels.
Kazakhstan Bureau of National Statistics, *21.2.a Air Quality*	Public statistical data include annual mean PM2.5 and PM10 concentrations in monitored cities. The indicator has a historical series starting in 2014.	Confirms the availability of official air-quality indicators that can support further analysis of the problem.
AirData.kz, *Open Air Quality Data for Kazakhstan*	The platform collects, processes, and publishes air-quality measurements from multiple sources in Kazakhstan, including historical observations and pollutant measurements.	Identifies a potential source of historical data for model development, subject to checking data coverage, completeness, variables, and suitability for the selected forecasting task.

2.3 Findings and Project Implications

The reviewed sources indicate that air pollution is a documented issue in Kazakhstan and that publicly accessible air-quality measurements are available for further analysis. Historical observations may provide a basis for investigating temporal patterns and developing a forecasting model.

These findings support the relevance of the proposed system, which aims to combine air-quality data processing, machine-learning forecasting, and a user-facing presentation of forecast results. The final dataset, forecasting target, and prediction horizon will be selected after assessing data suitability and availability.

This assessment is based on public-source analysis. No user interviews, client agreement, or independent statistical analysis of the datasets are claimed in this section.

3. Literature Review

The literature review draws on twelve sources published between 2021 and 2026. The studies cover IoT-based air-quality monitoring, forecasting methods, and integrated systems. The table summarizes their objectives, data, methods, findings, limitations, and relevance to the proposed project.

Source / study	Objective	Data	Methods	Finding / limitation	Relevance
Moursi et al. (2021), IoT-enabled monitoring and prediction on the edge	IoT monitoring and edge forecasting	PM2.5; sensor/edge data	NARX hybrid, LSTM, XGBRF	Combines IoT acquisition with edge/cloud prediction; transferability depends on setup	Supports modular IoT and forecasting architecture
Intelligent Forecasting of Air Quality and Pollution Prediction (2022)	Compare ML approaches	PM2.5 data	Linear Regression, RF, KNN, Ridge, Lasso, XGBoost, AdaBoost	Results depend on dataset and setting	Supports baseline-versus-model comparison
Jabbar et al. (2022), LoRaWAN-based outdoor monitoring	Long-range outdoor monitoring	Pollutant and climate sensor measurements	Arduino, LoRaWAN, TTN, ThingSpeak, Virtuino	Primarily monitoring; validates sensor/network setup	Informs acquisition and communication choices
Machine learning algorithms to forecast air quality: a survey (2023)	Survey forecasting research	155 papers, 2011–2021	Survey/review	Maps forecasting approaches; coverage depends on included literature	Broad methodological context
AQI prediction using ML for Ahmedabad (2023)	AQI prediction for Ahmedabad	CPCB/SAFAR, 2015–Jan 2021	SARIMA, SVM kernels, LSTM	RBF SVM reported best among tested methods; city-specific	Supports comparison of statistical, SVM, and recurrent methods
Prediction of AQI Using ML Techniques: Comparative Analysis (2023)	Compare AQI prediction methods	City-specific AQI datasets	SVR, Random Forest Regression, CatBoost; SMOTE discussed	Results depend on dataset and preprocessing	Supports careful comparison of regression models
Attention hybrid deep learning and quantum-inspired optimization (2024)	Improve AQI forecasting	Seoul data	ARIMA, attention CNN, LSTM, QPSO, XGBoost	Advanced hybrid; complexity may exceed prototype needs	Advanced reference, not default implementation
Supervised ML Approaches for Predicting Key Pollutants: Systematic Review (2024)	Review supervised ML for pollutants	Studies selected through PRISMA	Systematic review	Summarizes supervised ML; conclusions depend on search scope	Supports literature synthesis
Air Quality Forecasting Using ML: Comparative Analysis and Ensemble Strategies (2025)	Compare models and ensembles	9,357 hourly records; PM2.5, NOx, CO, benzene	Multiple ML, Bayesian optimization, stacking	Data are from 2004–2005 despite 2025 publication	Useful for evaluation design; note data age
Yildiz & Sucuoglu (2025), Real-Time IoT-Based Air Quality Forecasting	Real-time low-cost IoT forecasting	Low-cost sensor and climate measurements	Raspberry Pi 5, preprocessing, cloud, GRU	Reports R^2 >0.93 and latency <130 ms in its setting; site-specific and calibration limitations	Closest end-to-end IoT + ML reference
Deep Learning-based Forecasting for Urban India (2026)	Urban forecasting with temporal neural models	Hourly Indian-city data, 2015–2020	LSTM, CNN-LSTM, Transformer	Geographic and dataset-specific generalization needs assessment	Informs optional deep-learning alternatives
China major cities AQI forecasting using integrative ML (2026)	AQI forecasting across Chinese cities	Beijing, Wuhan, Xi'an	OMVMD–WLSSVM–GRM; LGBM/SATLDE	Complex optimized hybrid approach	Shows advanced options but not default implementation

Critical synthesis and preliminary research gap

The literature covers three connected but distinct areas: (1) IoT data acquisition and monitoring, (2) forecasting with statistical and ML methods, and (3) end-to-end systems combining sensing, preprocessing, prediction, and presentation. Monitoring-only work informs acquisition and communication design but should not be described as forecasting. Forecasting studies range from regression and tree ensembles to recurrent and hybrid deep-learning models; results are not directly comparable unless data, horizon, and metrics are aligned.

A defensible project direction is a manageable and reproducible prototype: a clear baseline, one or more candidate ML models, consistent evaluation conditions, and an IoT-compatible data input design. State a project-specific research gap cautiously; do not claim that no existing system combines IoT and ML, since the reviewed literature includes end-to-end examples.

4. Existing System Comparison

The table compares four systems described in the reviewed papers. "Not specified" means that the source reviewed for this comparison did not provide enough information to confirm the feature.

Criterion	A. Moursi et al. (2021)	B. Jabbar et al. (2022)	C. Yildiz & Sucuoglu (2025)	D. Urban India temporal DL (2026)
Primary purpose	Monitoring and forecasting	Outdoor monitoring / long-range transfer	Real-time IoT monitoring and forecasting	Air-quality forecasting
Physical IoT sensing	Yes	Yes	Yes	Not specified
Forecasting	Yes	Not central	Yes	Yes
Edge processing	Edge/cloud	Not specified	Raspberry Pi 5 edge layer	Not specified
Cloud/platform	Edge/cloud	TTN and visualization platforms	Cloud transfer, storage, visualization	Not specified
ML methods	NARX hybrid, LSTM, XGBRF	Not central	GRU and other compared models	LSTM, CNN-LSTM, Transformer
Visualization	System-dependent	ThingSpeak/Virtuino	Cloud visualization	Not specified
Evaluation	Prediction evaluation	Measurement/monitoring validation	Forecast metrics and latency	Forecast model comparison

Comparative interpretation

- Moursi et al. (2021) integrates IoT data with edge/cloud prediction and is an architectural reference.
- Jabbar et al. (2022) demonstrates sensor acquisition and LoRaWAN communication, but is primarily a monitoring system.
- Yildiz and Sucuoglu (2025) is the closest end-to-end reference in this set; its results are tied to its hardware, site, and evaluation conditions.
- The 2026 urban India study informs temporal-model comparison, but the available summary does not establish a physical IoT sensing pipeline.

Implication for the proposed project

The project should combine the capabilities required by its scope—data input, preprocessing, storage, forecasting, presentation, and measurable evaluation—without claiming the combination is unprecedented. Its contribution should be demonstrated through implementation, reproducible model comparison, and clear limitations.

5. Technology Comparison

The options below are based on the reviewed literature. Not every option was compared directly in the cited studies.

Decision	Alternatives	Advantages	Trade-offs	Preliminary direction
Data acquisition	Physical sensors; historical/public datasets; hybrid AirData.kz Open Air Quality Dataset for Kazakhstan Beijing Multi-Site Air Quality Dataset (UCI)	Sensors support live data; datasets support repeatable experiments	Hardware adds cost/calibration; datasets may not represent live local conditions	Start with historical data and an IoT-compatible input interface; add sensors only if available Evaluate AirData.kz as the primary candidate for Kazakhstan-focused forecasting and use the UCI Beijing dataset as a separate alternative for methodological experiments if needed.
Data transfer	Wi-Fi; LoRaWAN; dataset/API ingestion	Wi-Fi is straightforward; LoRaWAN supports long-range low-power networks; APIs/datasets are reproducible	Wi-Fi needs network; LoRaWAN needs compatible hardware/gateway; API availability varies	Use dataset/API initially; keep interface modular
Forecasting	Regression/SARIMA; RF/XGBoost; LSTM/GRU	Baseline is useful for comparison; tree models suit tabular features; recurrent models capture temporal patterns	Performance depends on data/horizon; deep learning needs more tuning and data	Compare baseline with at least one tree model; add GRU/LSTM only if justified
Storage	SQLite; PostgreSQL; cloud IoT/data platform	SQLite is simple locally; PostgreSQL supports structured backend; cloud platforms simplify some services	SQLite has concurrency limits; PostgreSQL needs setup; cloud adds configuration/dependency	SQLite for minimal local prototype or PostgreSQL for separate backend; decide after architecture
Processing/deployment	Server/cloud; edge; hybrid	Server-side is simpler to develop; edge can reduce latency; hybrid distributes work	Edge has resource constraints; cloud needs connectivity; hybrid adds complexity	Start with server-side forecasting and modular components

Technology inventory across the reviewed papers

Category	Technologies / methods identified
Acquisition and sensing	Physical IoT sensors; PM2.5 and other pollutant measurements; temperature/humidity; historical datasets
Communication / IoT platforms	LoRaWAN; Wi-Fi; Arduino Uno; Raspberry Pi; The Things Network (TTN); ThingSpeak; Virtuino
Statistical / baseline	Linear Regression; Ridge; Lasso; ARIMA; SARIMA
Tree-based and ensemble ML	Random Forest; XGBoost; AdaBoost; CatBoost; LightGBM; stacking
Other classical ML	KNN; SVM/SVR; Bayesian Ridge; Elastic Net; Decision Tree
Deep learning	LSTM; GRU; CNN-LSTM; CNN-GRU; Transformer; Attention CNN
Hybrid / optimization	NARX hybrids; ARIMA + attention CNN + QPSO-LSTM + XGBoost; OMVMD–WLSSVM–GRM with LGBM/SATLDE
Evaluation metrics	MAE; MSE; RMSE; R^2; classification accuracy; inference latency

The table lists methods found in the papers. It does not mean that every method will be implemented. The final selection should be small enough to evaluate under the same conditions.

6. Requirements Specification

The requirements below define the initial scope and acceptance criteria. The forecast target, horizon, and final dataset remain subject to confirmation based on data availability and supervisor feedback.

ID	Requirement	Type / priority	Acceptance criterion
FR-01	Accept air-quality records from a supported dataset/input source	Functional / Must	A sample dataset loads and fields are checked
FR-02	Preprocess input data before forecasting	Functional / Must	Missing values, types, timestamps, and features are handled by a documented pipeline
FR-03	Generate a forecast for a defined target and horizon	Functional / Must	Valid input produces a forecast; target and horizon are documented
FR-04	Provide a baseline forecast	Functional / Must	Baseline and ML results use the same test period
FR-05	Display or export forecast results	Functional / Must	User can inspect/retrieve forecast values and timestamps
FR-06	Calculate evaluation metrics	Functional / Must	MAE and RMSE are calculated; R^2 used where suitable
FR-07	Support future IoT integration through a modular input interface	Functional / Should	Ingestion interface is documented and separated from model logic
NFR-01	Avoid temporal data leakage	Non-functional / Must	Time-series data use chronological split; preprocessing fits training data only
NFR-02	Make results reproducible	Non-functional / Must	Dataset, preprocessing, split, settings, and metrics are documented
NFR-03	Keep the prototype modular	Non-functional / Should	Ingestion, preprocessing, forecasting, and presentation are separate components
NFR-04	Communicate limitations	Non-functional / Should	Documentation states coverage, horizon, limitations, and intended use

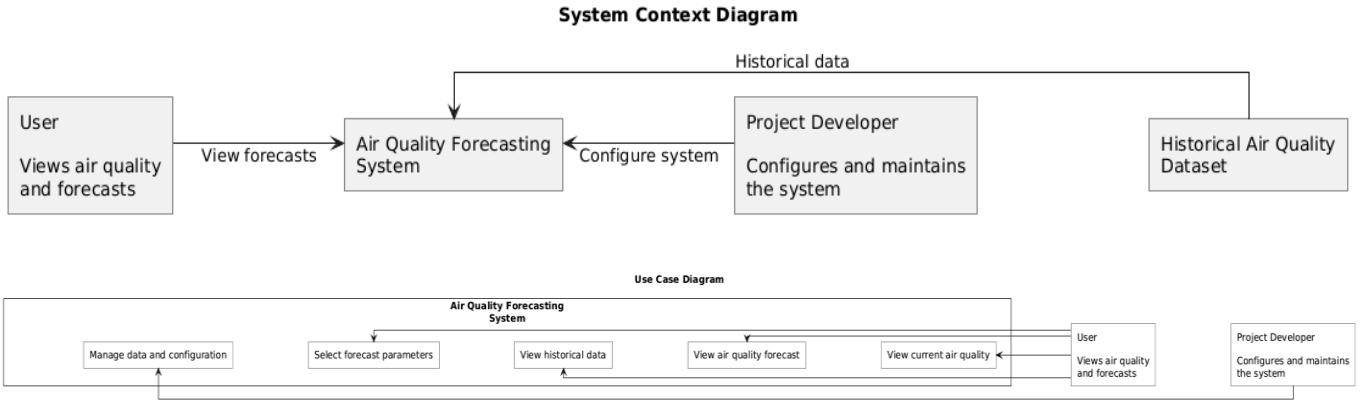
7. Architecture and Diagrams

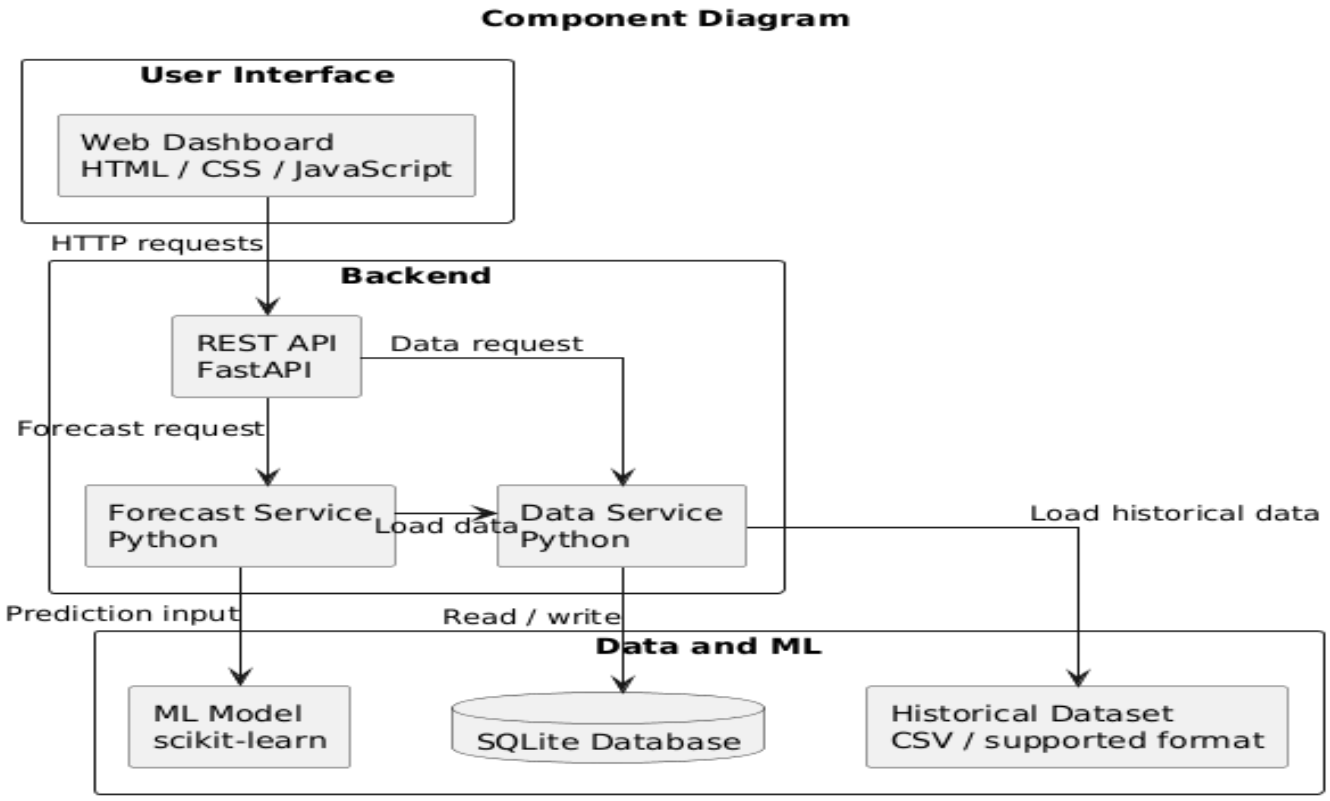
The System Context, Use Case, and Component diagrams have been prepared and are included below. The table summarizes the main elements represented in each diagram.

Diagram	Minimum content
System Context Diagram	User; forecasting system; dataset/API or IoT source; external storage/service if used; inputs and outputs
Use Case Diagram	Load/select data; preprocess; train/evaluate model; request/view forecast; inspect metrics
Component Diagram (draft)	Ingestion; validation/preprocessing; storage; baseline/model training and inference; evaluation; presentation/API

Draft component flow

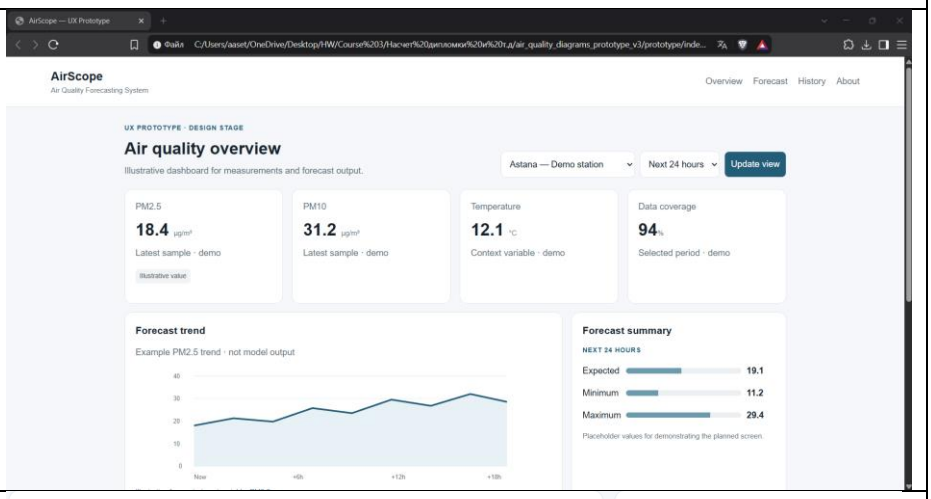
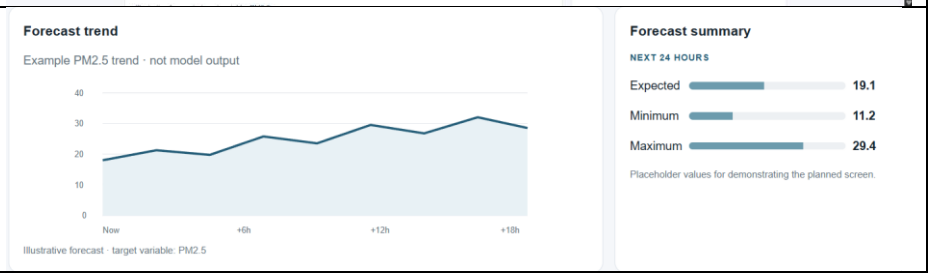
Data source → ingestion and validation → preprocessing/features → storage → baseline and ML forecasting → evaluation → forecast presentation/export.





8. UX / Prototype

An initial interactive HTML dashboard prototype has been prepared. The screenshots below show the Overview and Forecast Results views. All displayed measurements and forecast values are illustrative mock data; no trained model, live sensor feed, or production API is connected.

Screen / interaction	Suggested contents	Evidence provided
Overview	Selected data source, latest available measurement, forecast target and horizon	
Forecast results	Timestamped forecast values, units, selected model	

All values shown in the prototype are illustrative mock data and do not represent actual measurements or model predictions.

9. Verification and Evaluation Plan

The evaluation plan defines the proposed validation process and performance measures. The final target and applicable metrics will be confirmed after the dataset is selected and the forecasting task is specified.

Area	Method	Measure / pass condition
Input validation	Test valid records, missing fields, invalid types, timestamp errors	Invalid records handled predictably; errors documented
Preprocessing	Check missing-value handling, feature processing, and chronological order	Valid features produced without future information leakage
Forecast quality	Compare baseline and ML model on same chronological hold-out period	Report MAE/RMSE and R^2 where appropriate; select based on measured results
Classification (if used)	Compare predicted and true AQI categories	Report accuracy and, if useful, precision/recall or F1
Runtime	Measure inference time on target environment	Report observed latency; define threshold after environment is known
Reproducibility	Repeat evaluation using recorded settings	Same split/configuration reproduces results within expected tolerance

Baseline and evaluation conditions

- Use chronological train/validation/test splits for time-series data; do not mix future records into training.
- Compare models on the same target, horizon, test period, and metrics.
- Document data source, geographic coverage, date range, sampling interval, missing-data handling, and features.
- Do not promise accuracy before experiments; report measured results and limitations
- Record the selected dataset, geographical coverage, observation period, sampling interval, available features, missing-data rate, and applicable usage terms before model training.

10. Project Management and Risks

10.1 Repository and traceability

- Maintain one GitHub or GitLab repository and give the supervisor/committee appropriate access.
- Use separate accounts and identifiable commit author details for both students; no shared account.
- Include README, documentation, editable diagrams, code, tests, configuration examples, and .gitignore.
- Never commit passwords, API keys, personal data, or confidential information.
- Create YouTrack tasks with owner, outcome, acceptance criteria, and deadline; link task IDs to branches, commits, and pull/merge requests.
- Use task/feature branches and review significant changes before merging.
- Tag each checkpoint (pre-defense-1, pre-defense-2, final-pre-defense) and preserve earlier tags.
- **Project repository:** https://github.com/Set001YT/air-quality-forecasting-iot-ml
- **Project management tool:** https://air-quality-forecasting.youtrack.cloud

10.2 Initial responsibility split

Workstream	Asset Iglikov	Iskander Ismagulov	Shared review
Project proposal and scope	Lead: problem, aim, objectives, target users, scope	Review feasibility and data/IoT assumptions	Agree final scope and terminology
Literature review	Lead: organize source matrix and critical synthesis	Cross-check technical details and relevance	Verify claims and citation consistency against the 12 supplied papers
Existing systems and technology	Review implications for ML evaluation	Lead: compare 3–5 systems and alternatives for acquisition, transfer, storage, and processing	Keep criteria consistent and distinguish paper findings from project choices
Requirements and diagrams	Lead: forecasting requirements, metrics, System Context and Use Case drafts	Lead: IoT/data flow and Component Diagram draft	Align requirements, components, and use cases
Dataset and ML	Define evaluation logic and leakage-prevention criteria	Lead: inspect dataset options and outline baseline/model experiment	Choose target, horizon, split, and metrics before claiming performance
UX / prototype and API	Review forecast outputs and usability requirements	Lead: draft API/input-output example and proposed interface flow	Use clearly labelled mock data until real outputs exist
Repository and tracking	Commit owned documentation and maintain identifiable commits	Coordinate repository/YouTrack setup and task structure	Assign owner, deliverable, acceptance criteria, and deadline to each task

10.3 Initial risks

Risk	Impact	Mitigation
Suitable local/forecast dataset unavailable	Model evaluation delayed	Select and document a historical dataset early; do not assume live sensor access
Sensors or gateway unavailable	Live IoT integration may not be possible	Keep ingestion modular; treat hardware integration as conditional
Too many model options	Prototype/documentation may not finish	Use a baseline plus one or two candidates
Temporal data leakage	Misleading performance	Use chronological splits; fit preprocessing only on training data
Contributions not traceable	Repository assessment may suffer	Use individual accounts, task-linked commits, reviews, and regular updates
Late supervisor feedback	Rework compressed	Share focused deliverables at checkpoints and record changes

11. Supervisor Feedback

The table below records the supervisor's feedback and the changes made in response.

Date	Deliverable / topic	Feedback / decision	Action	Evidence / link
19 October 2026	Evidence of relevance, diagrams, prototype	These sections are urgent and should be completed for the checkpoint.	Prepare the user-discussion evidence, required diagrams, and a simple prototype.	Added
19 October 2026	Literature review	The review may include additional sources. Sources in Russian can be used, with relevant information written in English.	Check the existing papers and add suitable sources if needed.	Added
19 October 2026	Technology stack	The planned stack in the project justification is confirmed.	Keep the project documents consistent with the agreed stack.	Project justification
19 October 2026	Remaining sections and dataset	The supervisor will review the other sections and provide specific comments. A suitable dataset still needs to be selected.	Wait for section-specific feedback; compare dataset options before confirming the choice.	Added

For each meeting, record the feedback, the action taken, and where the updated material can be found.

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